

CEYLON GEM ARCHIVE

Project Plan & Implementation Roadmap

A Complete Gemstone Business Operating System
Built for Serendib Gemstones

Client	Serendib Gemstones
Platform	ceylon-gem-archive.vercel.app
Version	1.0 (Current)
Date	June 2026
Tech Stack	React + TypeScript + Tailwind + Supabase + Vercel

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1. PROJECT OVERVIEW

Ceylon Gem Archive (CGA) is a web-based gemstone business operating system designed specifically for colored gemstone dealers. Unlike general jewelry or diamond-focused inventory systems, CGA is purpose-built for the unique workflows of Sri Lankan gem traders dealing in sapphires, rubies, and other precious colored stones.

The system serves as a comprehensive platform for managing gemstone inventory, tracking costs and provenance, sharing stones with potential buyers through secure links, and maintaining complete documentation for each stone in the collection.

Project Goals:

- Replace spreadsheet-based gemstone tracking with a professional digital system
- Provide secure, beautiful buyer presentation links for international clients
- Track complete stone journey from acquisition through sale
- Maintain full cost breakdown and profitability analysis per stone
- Build a competitive advantage over other gem dealers through technology

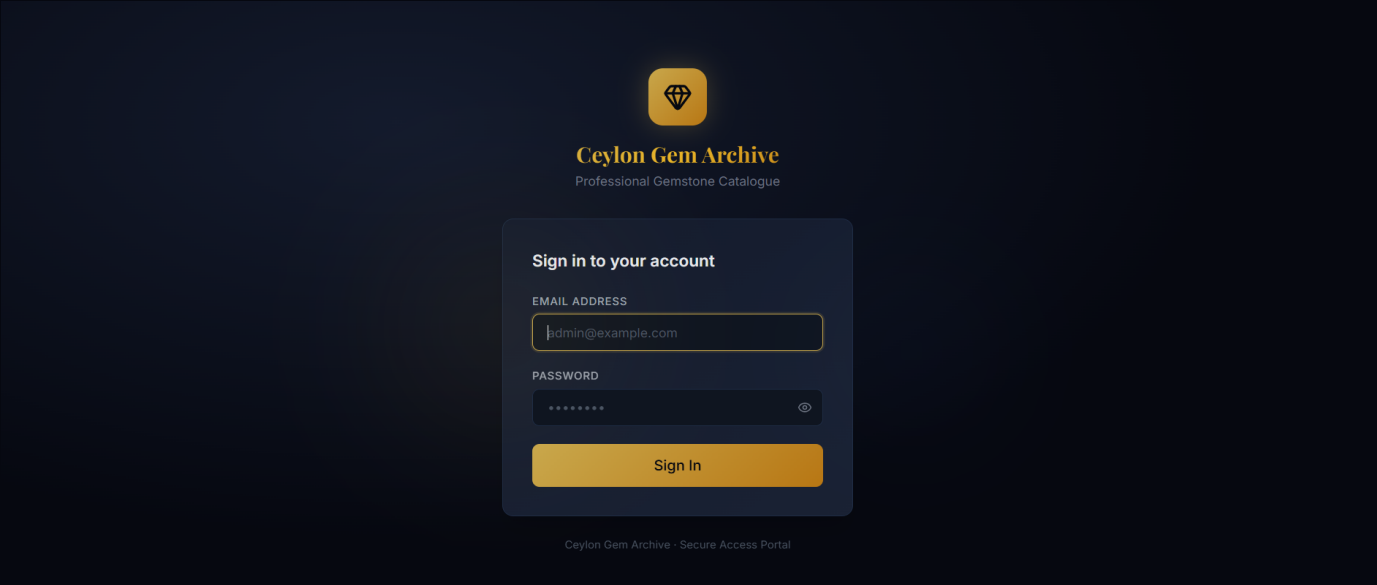


Fig 1. Secure login portal with branded design

2. CURRENT FEATURES (BUILT & WORKING)

Dashboard & Navigation

- ✓ Overview dashboard with gemstone statistics and quick actions
- ✓ Recent stones quick access
- ✓ Responsive sidebar with mobile drawer support
- ✓ Grid and list views for gemstone archive
- ✓ Search and filter functionality

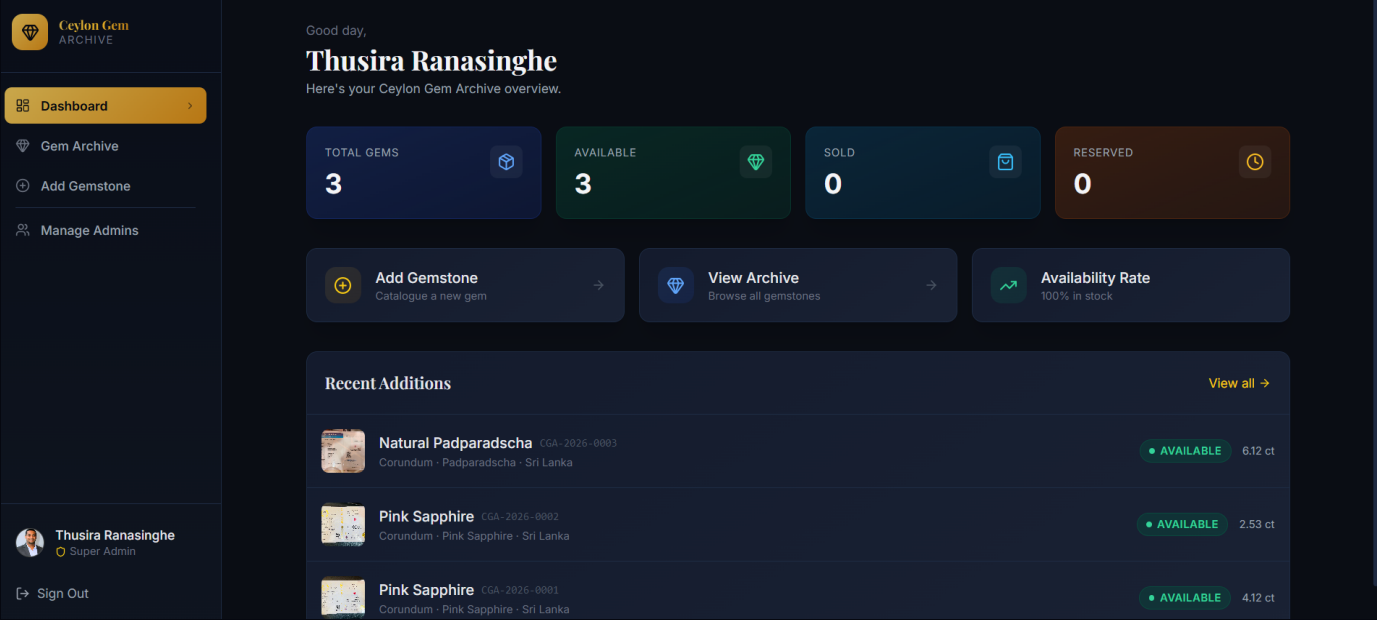


Fig 2. Dashboard overview with stats, quick actions, and recent additions

Gemstone Database & Archive

- ✓ Complete stone profiles: species, variety, weight, dimensions, color, clarity, cut, treatment, origin
- ✓ Auto-generated archive numbers (CGA-YYYY-NNNN) for unique identification
- ✓ Stone state tracking (Rough / Cut & Polished)
- ✓ Status management: Available, Reserved, Sold, Exported

Gem Archive
3 gemstones catalogued

[Add Gemstone](#)

Search by name, archive number, origin, species...

All Status

ARCHIVE #	GEMSTONE	ORIGIN	WEIGHT	STATUS	PRICE	ADDED BY	ACTIONS
CGA-2026-0003	Natural Padparadscha Corundum · Padparadscha	Sri Lanka	6.12 ct	• AVAILABLE	—	Thusira Ranasinghe	View Edit Share
CGA-2026-0002	Pink Sapphire Corundum · Pink Sapphire	Sri Lanka	2.53 ct	• AVAILABLE	\$1,800,000	Thusira Ranasinghe	View Edit Share
CGA-2026-0001	Pink Sapphire Corundum · Pink Sapphire	Sri Lanka	4.12 ct	• AVAILABLE	—	Thusira Ranasinghe	View Edit Share

Fig 3. Gem Archive list view with search, filters, and quick actions

Gemstone Detail View

- ✓ Rich detail page with photo/video gallery and all stone properties
- ✓ Share, Edit, and Delete actions from the detail header
- ✓ Video playback with navigation arrows and drag-to-reorder thumbnails

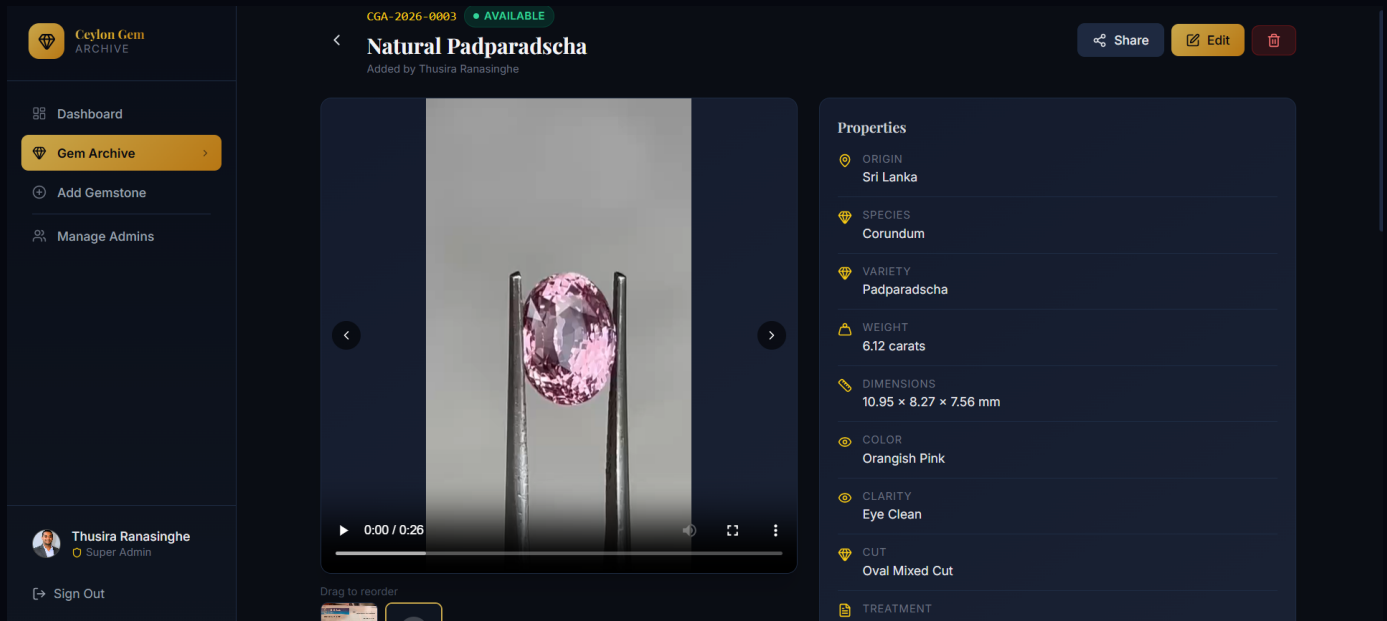


Fig 4. Gemstone detail page showing video gallery and full property panel

Add & Edit Gemstone Form

- ✓ Comprehensive form: Basic Information, Physical Properties, Certification
- ✓ Dropdown selectors for species, origin, clarity, and status
- ✓ Auto-generated archive numbers on creation

Cost Breakdown & Pricing

- ✓ Complete cost tracking: buying price, treatment, certification, other costs
- ✓ Support for Rough and Cut & Polished cost structures
- ✓ 10 currencies supported with live exchange rate conversion
- ✓ Selling price, buyer price, and sold price with separate currencies
- ✓ Profit margin calculation with percentage display
- ✓ Receipt and bill uploads linked to cost records

Ceylon Gem
ARCHIVE

Dashboard

Gem Archive

Add Gemstone

Manage Admins

Thusira Ranasinghe

Super Admin

Sign Out

Certification

CERTIFICATE NUMBER

CERTIFYING LABORATORY

e.g. GIA-1234567

None

Cost Breakdown

All costs are internal only and never shown to buyers.

STATE OF STONE WHEN PURCHASED

Rough

Cut & Polished

CURRENCY

USD (\$)

ROUGH STONE PRICE

ROUGH STONE WEIGHT (CT)

\$ 0.00

e.g. 12.5

PREFORMING COST

CUT & POLISH COST

\$ 0.00

\$ 0.00

TREATMENT COST

CERTIFICATION COST

OTHER COSTS

\$ 0.00

\$ 0.00

\$ 0.00

Transport, packaging, etc.

Fig 5. Cost Breakdown section with Rough/Cut & Polished toggle and multi-currency support

Ceylon Gem
ARCHIVE

Dashboard

Gem Archive

Add Gemstone

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Super Admin

Sign Out

Selling Prices

Profit margins are calculated against total purchase cost.

MIN. SELLING PRICE

BUYER PRICE

PRICE SOLD

USD (\$)

USD (\$)

USD (\$)

\$ 0.00

\$ 0.00

\$ 0.00

Internal only

Shown to buyers

Final sale price

Notes

INTERNAL NOTES

Never shown to buyers

Internal observations, provenance notes, purchase history...

BUYER NOTES

Shown on buyer share pages

Description for potential buyers...

Fig 6. Selling Prices with internal/buyer/sold pricing and Notes section

Certificate Management

- ✓ Store certificate number and certifying laboratory per stone
- ✓ Support for all major labs: GIA, GRS, GGTL, SSEF, EGL, AIGS, Lotus Gemology
- ✓ Certificate display on buyer share pages

Provenance & History Timeline

- ✓ Visual timeline of stone journey events
- ✓ Track: event type, date, location, performed by, notes
- ✓ Inline editing of history entries from both detail and edit pages
- ✓ Add, edit, and delete events at any time

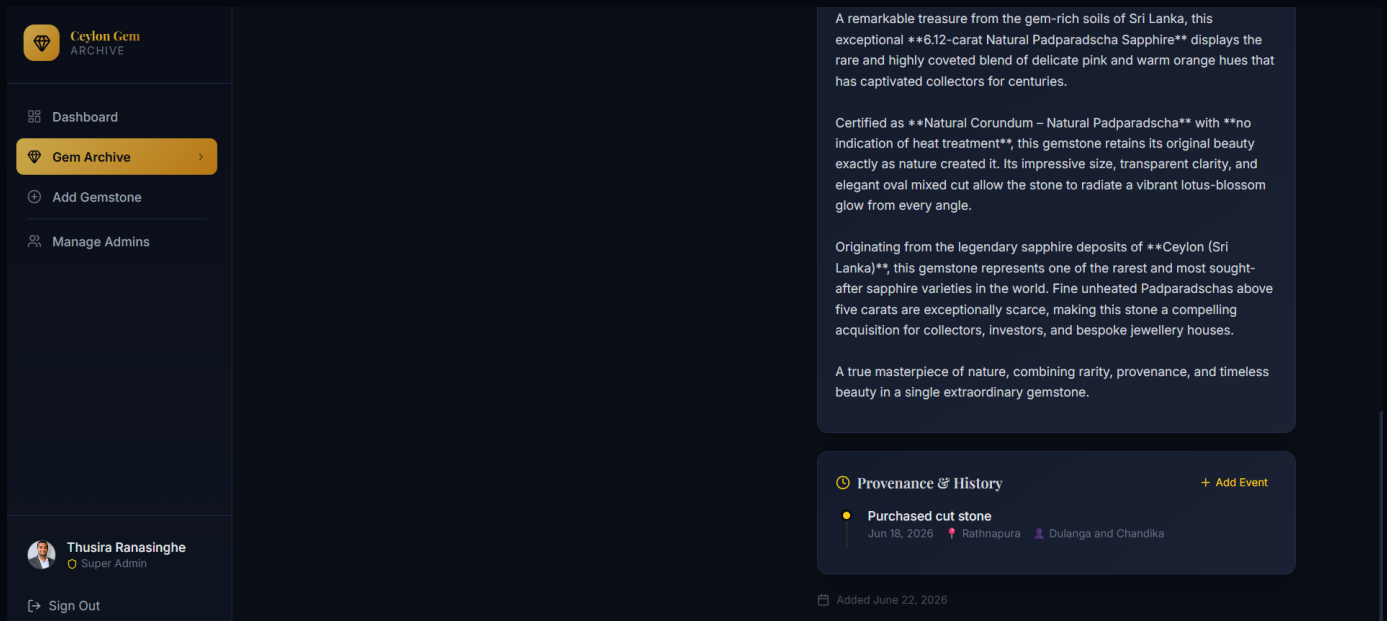


Fig 7. Buyer Notes display and Provenance & History timeline with event tracking

Photo & Video Management

- ✓ Multi-file upload with drag-and-drop (JPG, PNG, WebP, MP4)
- ✓ Image reordering with drag-to-sort functionality
- ✓ Primary image selection for thumbnails
- ✓ Full-screen lightbox viewer with image rotation and save
- ✓ Video playback with custom thumbnails

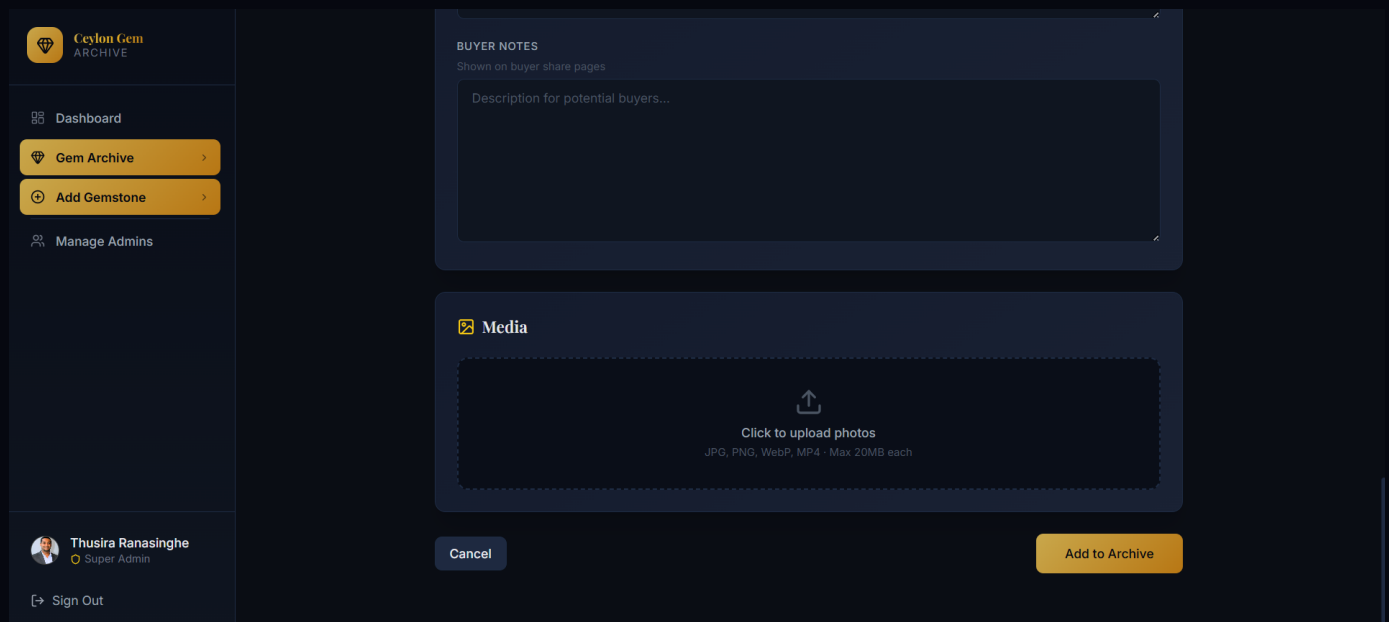


Fig 8. Media upload area with drag-and-drop support for photos and videos

Buyer Share Links

- ✓ Generate secure, token-based share links for any gemstone
- ✓ Configurable expiry dates: 1 hour to 90 days, or never
- ✓ View count tracking per share link
- ✓ Beautiful public presentation page (no login required)
- ✓ Representative contact info with profile photo, Call & WhatsApp buttons

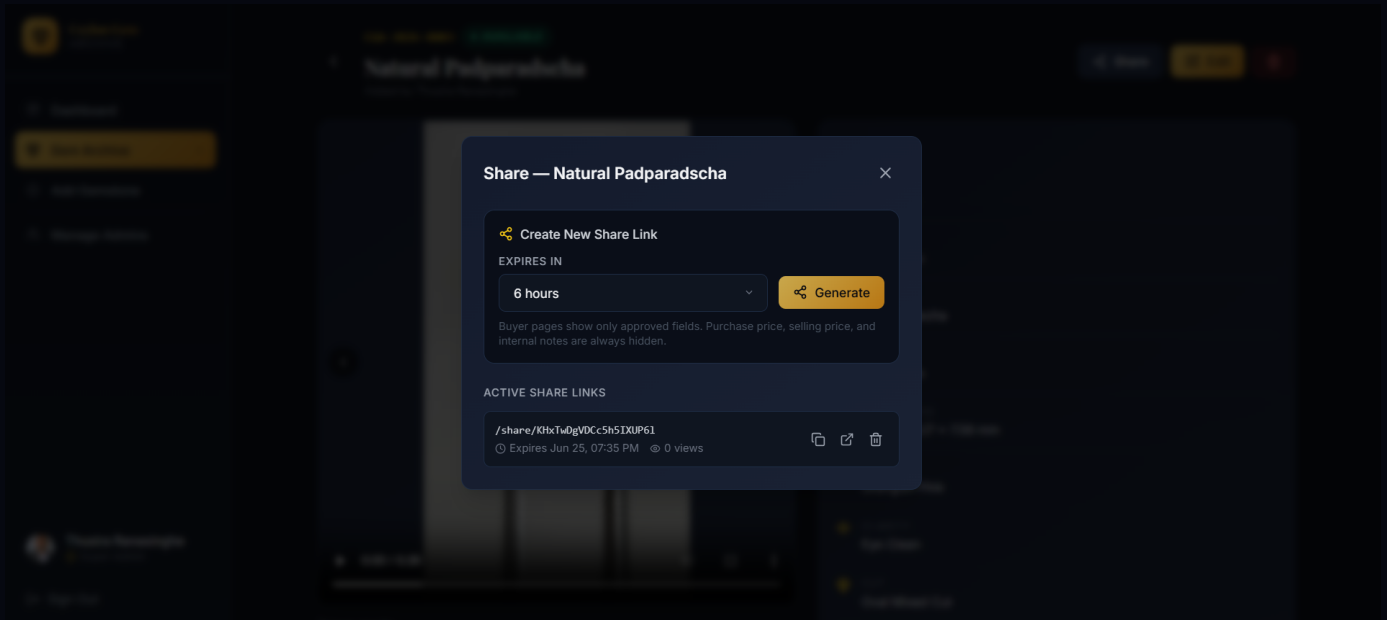


Fig 9. Share dialog with expiry settings, active link management, and view tracking

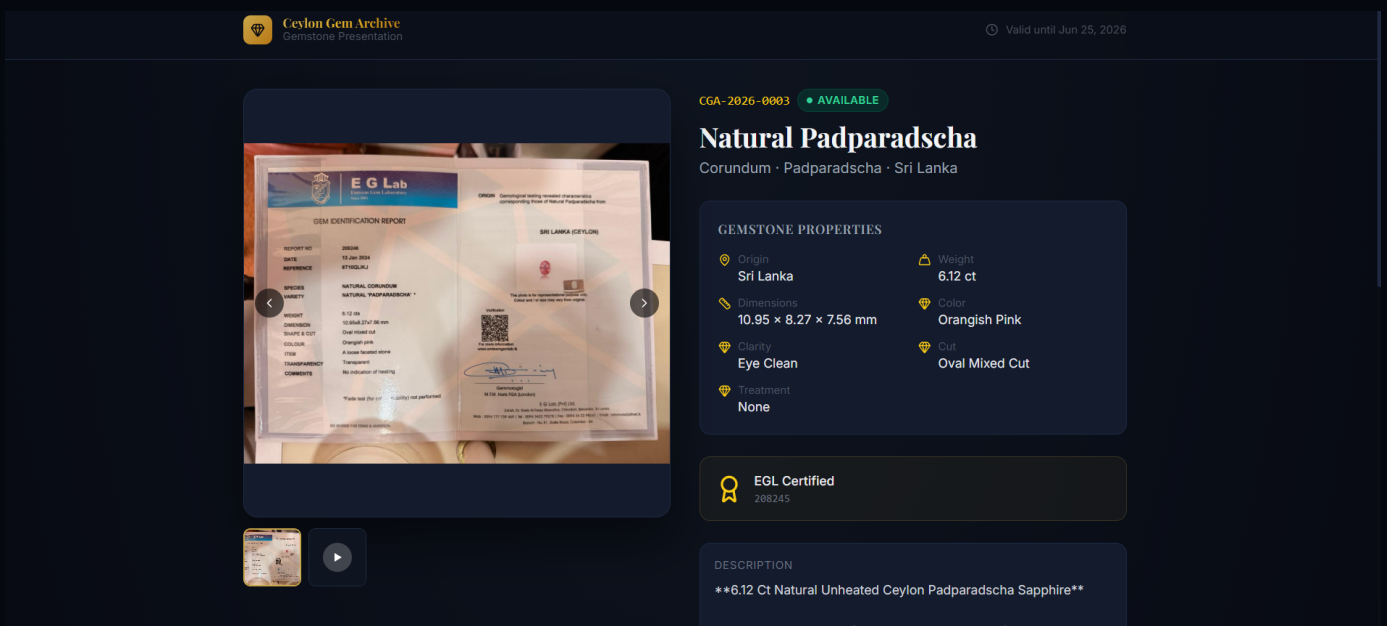


Fig 10. Buyer presentation page — public view with properties, certificate, and gallery

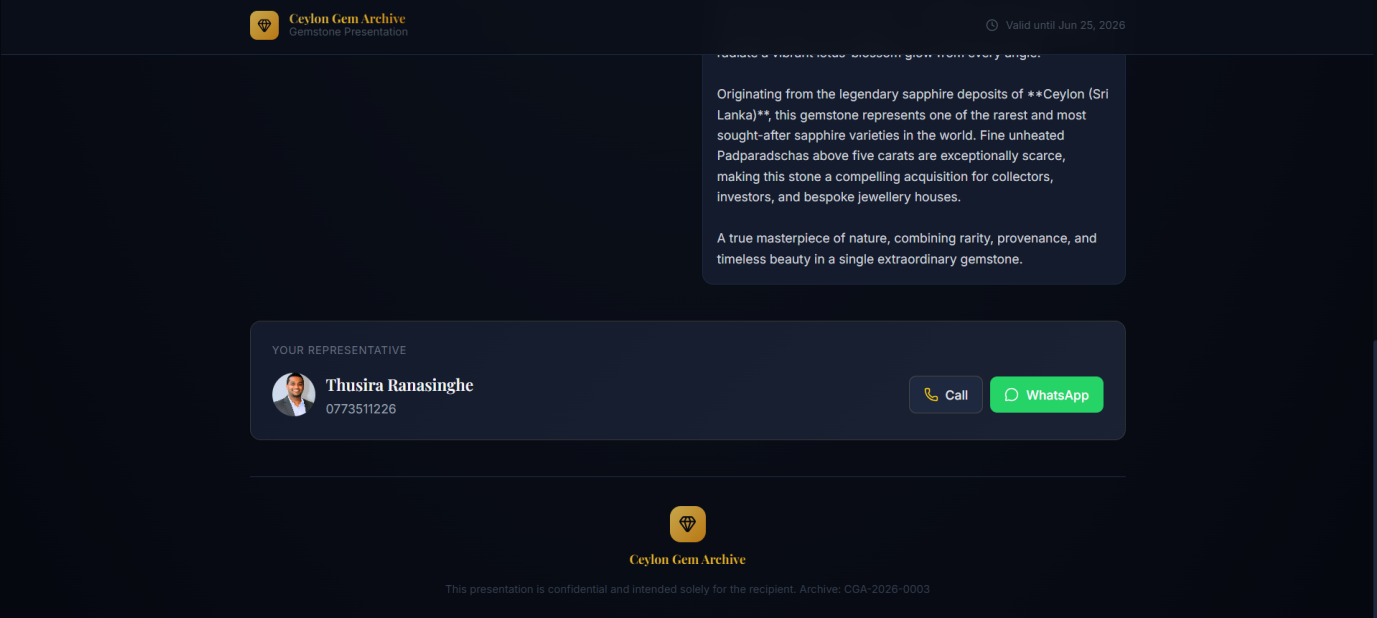


Fig 11. Buyer page footer with representative contact, Call & WhatsApp buttons

Authentication & Admin Management

- ✓ Secure Supabase authentication with email/password
- ✓ Role-based access: Super Admin and Admin roles
- ✓ Admin invite system with unique signup links
- ✓ Admin blocking, promotion, and management
- ✓ Profile management with avatar upload

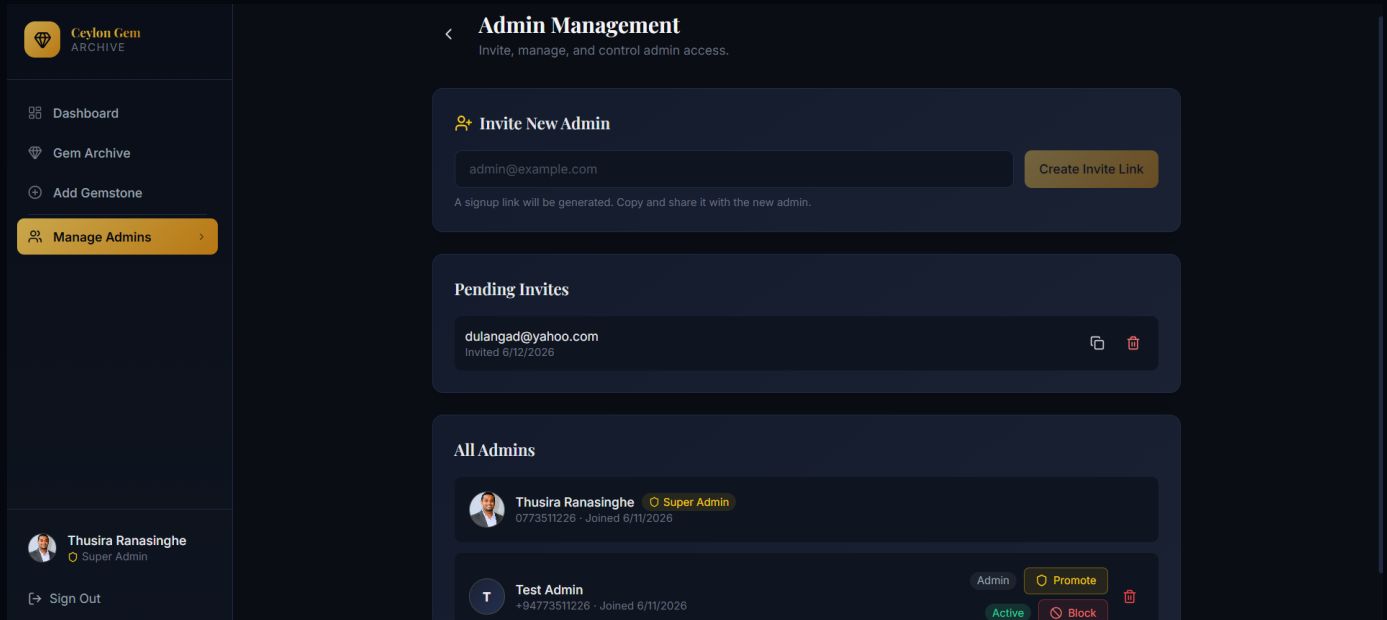


Fig 12. Admin Management — invite system, pending invites, role management

Notes System

- ✓ Internal notes (never shown to buyers)
- ✓ Buyer notes with paragraph formatting (shown on share pages)
- ✓ Resizable text areas for comfortable editing

3. TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	React + TypeScript	Component-based UI with type safety
Styling	Tailwind CSS	Utility-first CSS with custom gem theme
Backend	Supabase	Authentication, database, storage, real-time
Database	PostgreSQL	Relational data with row-level security
Storage	Supabase Storage	Photos, videos, certificates, receipts
Hosting	Vercel	Edge deployment with automatic CI/CD
Routing	React Router	Client-side navigation
Icons	Lucide React	Consistent icon system

Architecture Highlights:

- Row-Level Security (RLS) on all database tables for data protection
- Public share pages accessible without authentication
- Optimized media delivery through Supabase CDN
- Responsive design supporting desktop and mobile browsers

4. IMPLEMENTATION ROADMAP

PHASE 1 — Close Critical Gaps

Goal: Match core features of Nveloop GMS and GemCloud to be competitive.

Feature	Description	Effort
Consignment / Memo Tracking	Track stones sent on approval: who has it, due date, memo terms. New "On Memo" status with reminders.	2-3 sessions
QR Code System	Generate QR per stone linking to buyer share page. Print sheets for gem shows.	1 session
Invoicing	Generate PDF invoices with logo, stone details, certificate info. Track payment status.	2 sessions
Supplier Management	Track suppliers, link to purchases, view history and amounts owed/paid per supplier.	1-2 sessions

PHASE 2 — Gain Competitive Advantage

Goal: Surpass competitors with unique features they don't offer.

Feature	Description	Effort
Buyer CRM	Full contact management, purchase history, interest matching, auto-suggest stones to buyers.	2-3 sessions
AI Certificate OCR	Upload any certificate (GIA, GGTL, GRS, SSEF, EGL), auto-extract all fields. First to support ALL labs.	1-2 sessions
Consignment Accounting	Dashboard of memo values, who has what, overdue tracking, profit/loss per stone.	2 sessions
Multi-Currency Accounting	Track payments in different currencies, exchange rate at time of sale.	1-2 sessions
Digital Vault	Store export permits, invoices, lab reports permanently linked to each stone.	1-2 sessions
Inventory Valuation Dashboard	Total cost, market value, potential profit. Group by status, origin, species.	1-2 sessions

PHASE 3 — Market Leadership

Goal: Build features no competitor offers, establishing CGA as the industry leader.

Feature	Description	Effort
Exhibition Mode	Named collections for shows, PDF/QR catalogues, temporary buyer group share links.	2 sessions
B2B Marketplace	Public catalogue with inquiry button, no prices shown. Integrated natively.	3-4 sessions

Feature	Description	Effort
Gemstone Passport	Blockchain-verified ownership, QR reveals full provenance, certificates, sale history.	4-5 sessions
Mobile PWA	Progressive web app for gem shows and field buying. Offline capable.	1 session
Role-Based Visibility	Hide costs from "Sales" role. Fine-grained field-level access control.	1 session
Watermark Protection	Buyer name overlay on shared images. Discourage unauthorized distribution.	1 session
Price Intelligence	Price-per-carat trends from sales data. Historical charts by species.	1-2 sessions
Audit Logs	Track who viewed, edited, downloaded, shared each stone. Full accountability.	1-2 sessions

5. COMPETITIVE LANDSCAPE

The gemstone-specific software market is thin, with only 3-4 truly dedicated tools. Most competitors are either diamond-focused or general jewelry retail systems. This represents a significant opportunity for CGA.

System	Price/Month	Focus	Target Market
Nveloop GMS	\$50 - \$140	Colored gemstone dealer management	Gem merchants (closest competitor)
GemCloud	\$89.99	Inventory + B2B marketplace	Gemstone traders
Smartrade	Unknown	Trade management	Enterprise gem businesses
CrystalDesk	Unknown	Merchant CRM	Gem merchants
Gem Logic	\$299 - \$2,500	Full jewelry store management	Retail jewelers
Jewel360	Unknown	Jewelry retail POS (GIA integration)	Jewelry retailers
WingoldNext	Unknown	Jewelry ERP & manufacturing	Jewelry manufacturers
Ceylon Gem Archive	Free (self-hosted)	Colored gemstone business OS	Sri Lankan gem dealers

Key Competitor Limitations:

- Nveloop limits photos to 2-3 per stone depending on plan tier
- No competitor offers buyer share links with public presentation pages
- No competitor provides stone provenance/journey timeline tracking
- Most have dated, utilitarian interfaces lacking modern design
- None specifically target the Sri Lankan gemstone market
- AI-powered certificate extraction is not available in any competitor

6. CGA COMPETITIVE ADVANTAGES

Advantage	Details	vs Competitors
Buyer Share Links	Secure, expiring links with beautiful public presentation pages	No competitor offers this
Unlimited Media	Upload unlimited photos and videos per stone with full management	Nveloop limits to 2-3 per stone
Provenance Timeline	Visual journey tracking from acquisition through sale	Unique to CGA
Modern UI/UX	Dark luxury theme with responsive design and smooth interactions	Competitors look dated
Receipt Uploads	Bills and receipts linked directly to cost breakdown records	Not available elsewhere
Cost Effective	Free / self-hosted with no monthly subscription fees	Competitors charge \$50-\$300+/month
Sri Lankan Focus	Built specifically for Sri Lankan gem dealers and their workflows	No competitor targets this market

7. SUMMARY & NEXT STEPS

Ceylon Gem Archive is already a functional gemstone business management system with features that surpass competitors in key areas like buyer presentation, media management, and provenance tracking. The three-phase roadmap will systematically close gaps and build market-leading capabilities.

Current Status:

- ✓ Core platform built and deployed at ceylon-gem-archive.vercel.app
- ✓ 9 major feature modules operational
- ✓ Active use for gemstone inventory management

Immediate Next Steps (Phase 1):

- Implement Consignment/Memo tracking for stone approvals
- Add QR code generation for gem show presentations
- Build invoicing system with PDF export
- Create supplier management module

Strategic Vision:

With Phase 1 and Phase 2 complete, CGA will offer everything that Nveloop and GemCloud provide, plus buyer share links, unlimited media, provenance tracking, AI-powered certificate OCR, and a modern user experience — all at a fraction of the cost. This positions CGA as the most complete and cost-effective gemstone business platform in the market.

Ceylon Gem Archive — Serendib Gemstones

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